Predicting Linear Relational Decodability Without the Relation

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Abstract

Linear relational embeddings (LREs) approximate a language model's subject-to-object mapping with an affine function. Their faithfulness varies enormously across relations – near zero for PERSON_FATHER, at ceiling for COUNTRY_CAPITAL_CITY. We ask what determines that variation, and show it is substantially predictable from a readout that never sees the relation. We estimate a Jacobian lens from subject-to-subject differences under a single neutral prompt, at one fixed layer for every relation, using no relation template, no objects and no (subject, object) pairs and measure how highly it ranks the model's own answer. Across 28 relations of the LRE dataset on Olmo-3-7B this relation-agnostic score correlates with LRE faithfulness at $\rho = 0.66$ ($p = 1.4 \times 10^{-4}$), rising to 0.75 once object cardinality is partialled out. Neither cardinality nor the model's prior over the answer token accounts for the correlation, and the estimate is stable under leave-one-domain-out (0.57–0.70). This suggests that linear decodability may be largely a property of what a subject's representation already carries, rather than of the relation being decoded.

1 Introduction

Hernandez et al. (2024) showed that a language model's mapping from a subject to the object of a relation is often well approximated by an affine function of the subject's hidden state – a *linear relational embedding* (LRE). They also showed that this works far better for some relations than others. On Olmo-3-7B, we reproduce the same spread: an LRE fitted on COUNTRY_CAPITAL_CITY agrees with the model's own prediction 75% of the time while one fitted on PERSON_FATHER agrees 9% of the time.

Why? The natural reading is that some relations are "more linear" than others – that linearity is a property of the relation. We report evidence that it is better understood as a property of the *subject's representation*. Concretely, we show that one can predict how faithful an LRE will be *before fitting it*, and without using the relation at all.

Our diagnostic is a *Jacobian lens*: a readout that propagates a first-order approximation of the model's own computation from an intermediate layer to the output vocabulary. Lenses of this kind have recently been used to characterise what a model is internally disposed to say (Gurnee et al., 2026). Ours differs in what it is applied to: rather than a corpus-averaged Jacobian acting on raw activations, we evaluate the Jacobian at individual reference entities and apply it to *differences* between entity representations, under a single neutral prompt shared by every relation. The lens sees no relation template, no objects, and no (subject, object) pairs; the layer is fixed in advance rather than tuned per relation. We then ask how highly it ranks the model's own answer to the relation. That score correlates with LRE faithfulness across 28 relations at $\rho = 0.66$.

We make three contributions. (i) We introduce a relation-agnostic diagnostic of linear decodability and show it predicts LRE faithfulness. (ii) We show the prediction survives controls for object cardinality and answer-token frequency, and for holding out entire subject domains. (iii) We document a first-token scoring artifact that silently inflates faithfulness to a perfect score on relations with numeric objects, and propose a remedy.

2 Background

An LRE for relation r approximates the model's computation as

$$o \approx \beta W s + b, \tag{1}$$

where s is the hidden state at the subject's final token, o the output state, and W, b are estimated from a small set of reference subjects as the mean Jacobian and mean bias of the model's own map (Hernandez et al., 2024).

The scalar β corrects a systematic weakness of the first-order estimate. A Jacobian taken at a reference subject describes the model's map only in a neighbourhood of that subject; extrapolated to a different subject it points in roughly the right direction but is too short. This matters because the prediction decomposes into a subject-dependent term and a constant: expanding b gives

$$\beta W s + b \;=\; \mathrm{mean}_i[\, f(s_i) + \beta J_i(s - s_i)\,], \quad (2)$$

in which only the second term depends on which subject is being decoded. Left unscaled, the shared anchor $\mathrm{mean}_i\, f(s_i)$ dominates and the operator returns much the same object for every subject. Setting $\beta > 1$ amplifies the subject-specific contribution relative to that anchor, and so does change which object is ranked first. Hernandez et al. (2024) fit β per relation alongside the layer; we follow their protocol and select it from $\{0.5, 1, 1.5, 2, 3, 5\}$. Eq. 2 is also what makes the LRE affordable: each test subject costs n Jacobian–vector products rather than a materialised 4096×4096 matrix.

Faithfulness is the rate at which decoding $\beta W s + b$ over the full vocabulary reproduces the *model's own* top-1 prediction – not the gold object. It is therefore a measure of how well an affine map captures what the model does. Faithfulness defined this way is the quantity we predict.

3 A relation-agnostic Jacobian lens

For an entity e (e.g., France) and a set of reference entities $\{i\}$ of the same type (e.g., Spain, Italy), we take the Jacobian–vector product of the model's map at each reference and average:

$$v_e \;=\; \frac{1}{n} \sum_i J_i\,(s_e - s_i). \quad (3)$$

Here J_i is the Jacobian of the model's map from the subject's hidden state at layer ℓ to the final-layer state at the last position, evaluated at reference i. We never form J_i explicitly: a Jacobian–vector product $J_i u$ can be obtained for the cost of a constant number of forward and backward passes,[1] which is what makes Eq. 3 affordable at $d_{\text{model}} = 4096$. Each J_i is evaluated under a *neutral* prompt, `"{} is known for"`, identical for every relation.

[1]We use the standard double-backward construction: differentiating $\langle u', f(h)\rangle$ with respect to h with u' held as a variable, then differentiating the result with respect to u' in the direction u. This requires `float32` and eager attention, as fused attention kernels are not twice differentiable.

Why differences. The quantity we want is not what the model says about France, but what is *specific* to France among countries. A raw activation s_e is dominated by what all entities of its type share – the type itself, the syntactic role, the prompt – and a lens applied to it returns mostly that shared content. Subtracting a same-type reference cancels it, leaving the entity-specific residual, and the Jacobian then carries that residual forward to the vocabulary. The lens therefore reads what this entity, as opposed to others of its kind, makes the model disposed to say.

We note that this differencing is not itself new: it falls directly out of the LRE construction. Eq. 2 already expresses the LRE as a constant plus a sum of $J_i(s - s_i)$ terms, so Eq. 3 is that same quantity with the subject-independent anchor dropped and $\beta = 1$. What is new here is not the algebra but the use: we estimate the lens without any relation and ask whether it predicts an LRE that was fitted *with* one. This also distinguishes our lens from that of Gurnee et al. (2026), which averages a Jacobian over a corpus and applies it to raw activations.

We decode v_e over the full vocabulary and record the rank of m_e, the model's own top-1 answer for relation r. Writing $\mathrm{rk}_{v_e}(m_e)$ for that rank, the diagnostic for a relation is

$$A_r \;=\; -\,\mathrm{median}_e\, \log_{10} \mathrm{rk}_{v_e}(m_e), \quad (4)$$

so higher A_r means the lens agrees with the model more.

Three properties make A_r non-trivial as a predictor. The prompt is neutral and shared, so no relation template enters. The differences in Eq. 3 are subject-minus-subject – *France* − *Spain*, never *France* − *Paris* – so no relation-specific (subject, object) pair is used in estimation. The layer is not relation-specific either. It is fixed at 17 for all relations.[2]

Relation-specificity enters in exactly one place: the *target*. The model's top-1 is taken under the same 8-shot in-context prompt used to measure faithfulness, so the two quantities are commensurable.

[2]Layer 17 is roughly half-way through the network, the region in which such lenses are usable at all (Gurnee et al., 2026), and is the modal best layer for the LRE itself in our sweep (10 of 28 relations; the rest spread over 5, 11, 14, 20 and 23). We stress that a single layer is used for *every* relation, so this choice cannot manufacture a correlation across relations: it can only make the diagnostic uniformly better or uniformly worse. Tuning the layer per relation would score higher and mean less.

4 Experimental setup

Model. Olmo-3-7B, in `float32` with eager attention, for the reason given in footnote 1.

Relations. The LRE dataset (Hernandez et al., 2024) contains 47 relations. Our prompt presupposes an entity subject, so we fixed a subject-type filter in advance using the dataset's declared `domain_name` field: the 28 relations with named-entity subjects (*person*, *country*, *name*, *company*, *landmark*, *superhero*, *city*, *pokemon*, *product*, *star*). Excluded are relations over words as words (*verb*, *adjective*, *word*) and over abstracts (*occupation*, *task*, *characteristic*, *degree*), for which `"give is known for"` is a category error. No other relation was excluded.

Measurement. Both quantities are read off the same prompt, so that they are comparable. Following Hernandez et al. (2024), a relation is presented to the model *in context*: eight (subject, object) examples of that relation are concatenated, one per line, and the query subject is appended without its object, so that the model's next token is its answer. This is what defines "the model's own prediction" for both measures. The subjects of a relation are shuffled once and partitioned into three disjoint sets: eight in-context examples, eight reference subjects from which W and b are estimated, and the remainder as scored test subjects.

1 <<< hs: in our field, it is not customary to say something like "replacing the mixture of earlier protocols in our own logs". if it is a relatively minor not "preregistered" change, then it does not have to be declared. if it is major, then it belongs in limitations >>>

Faithfulness is then recomputed for all 28 relations under this protocol, with layer and β chosen on a 6×6 grid by the mean over the test subjects. The lens uses the same eight reference subjects and scores the same test subjects (11–20 per relation, depending on how many remain after the in-context and reference sets are removed). Crucially, the in-context prompt is used only to obtain the *target*; the lens itself is always computed under the neutral prompt of §3.

5 Results

Figure 1 shows the result. The relation-agnostic score predicts LRE faithfulness at Spearman $\rho = 0.658$ ($p = 1.4 \times 10^{-4}$). With 28 points and a parametric p-value resting on assumptions we would rather not make, we also test the correlation by permutation: we shuffle the lens scores against the faithfulness values 10^4 times, recompute ρ each time, and count how often the shuffled $|\rho|$ matches or exceeds the observed one. It does so in 0.03% of shuffles, so the association is not an artifact of the small sample.

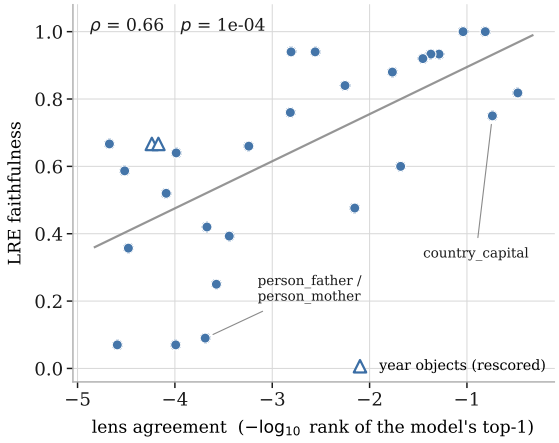


Figure 1: LRE faithfulness against the relation-agnostic lens score, 28 relations. The lens sees no relation template, no objects, and no (subject, object) pairs, and uses one fixed layer throughout. Triangles mark the two relations with numeric objects, rescored as described in §7.

The correlation is not carried by cardinality. Relations differ greatly in how many distinct objects they have; k, the mean number of subjects per object, ranges from 1.0 to 189. High k makes faithfulness easier, and indeed k predicts faithfulness on its own ($\rho = 0.43$, $p = 0.02$). But k is essentially uncorrelated with the lens score ($\rho = -0.03$, $p = 0.87$; that is, no detectable relationship, which is what we want here, though with $n = 28$ we can only rule out a strong one). Partialling k out *raises* the correlation to $\rho = 0.744$ ($p = 10^{-5}$): k and the lens are independent predictors of faithfulness, so removing the variance k explains sharpens rather than weakens the lens signal.

Nor by answer frequency. A relation whose answers are frequent tokens might be easy for both measures. We estimate the model's prior over each answer token by averaging its log-probability across ten neutral contexts. This predicts neither faithfulness ($\rho = -0.03$) nor the lens score ($\rho = -0.02$), and partialling it out leaves the cor-

relation unchanged at 0.658; with both covariates (cardinality and answer frequency), 0.746.

How we partial. All variables here are ordinal, so we work throughout on ranks. To partial out a set of covariates we rank-transform the lens score, faithfulness and each covariate; regress the ranked lens score on the ranked covariates by ordinary least squares with an intercept; do the same for ranked faithfulness; and report the Pearson correlation of the two residual vectors. This is the standard partial Spearman correlation, and it asks how much of the lens–faithfulness association survives once everything the covariates can linearly explain about either variable has been removed.

Nor by one domain. Eleven of the 28 relations have *person* subjects, raising the worry that one domain drives the result. Holding out each subject domain in turn leaves $\rho \in [0.570, 0.700]$, median 0.656.

A within-domain check. PERSON_FATHER and PERSON_MOTHER are separate relations over near-identical subject sets. They receive almost identical faithfulness (0.09, 0.07) and almost identical lens scores (-3.69, -3.99), without the lens being told which relation it is looking at.

6 Where the account degrades

What the lens reads is what an entity is *salient* for. On the account proposed here that is not a confound to be excused but the mechanism itself: a relation is linearly decodable to the extent that its answer is already part of what the subject's representation makes the model disposed to say. The relations that deviate from the trend are therefore exceptions to the account, not failures of measurement, and it is worth asking whether they are unstructured.

They are not. Ordering relations by their residual from the fitted trend, both tails are organised by the size of the answer space. The largest positive residuals – faithful despite a low lens score – are LANDMARK_ON_CONTINENT ($+0.30$, $k = 189$), NAME_GENDER ($+0.29$, $k = 9.5$) and PERSON_NATIVE_LANGUAGE ($+0.26$, $k = 30.6$). In each, the LRE chooses among a handful of possible answers, and can be faithful without the answer being salient for any particular subject: no first name is chiefly "known for" its gender, but there are only two answers to pick between. The largest negative residuals – unfaithful despite the lens – are PERSON_FATHER (-0.43), PERSON_MOTHER (-0.41) and COMPANY_CEO (-0.32), all with $k \approx 1$ and answer spaces in the hundreds, where faithfulness is near its floor whatever the representation carries.

The account thus holds in the middle of the cardinality range and degrades predictably at both ends, in opposite directions. This is the same variable whose removal raises the correlation to 0.744, and it gives that number an interpretation: partialling out k removes the principal source of deviation from the salience account rather than merely a nuisance correlate.

7 A first-token scoring artifact

Both relations with numeric objects (PRESIDENTS_BIRTH_YEAR, PRESIDENTS_ELECTION_YEAR) initially scored faithfulness 1.000. This was spurious. In this tokenizer `" 1743"` segments as `[' ', '174', '3']`: digits never merge with a preceding space. First-token scoring therefore compared the model's top-1 against a bare space token on both sides, and faithfulness 1.000 meant only that the LRE and the model both emit a space.

The fix is to place the space in the prompt, so that the query ends with it and the next token is content. Rescored, the two relations give faithfulness 0.67 and lens scores of -4.24 and -4.17; the model does predict the year correctly once the space is consumed. Including the uncorrected values would have depressed the reported correlation from 0.658 to 0.427.

We flag this because the failure is silent: it produces a perfect score rather than an obvious error, and it will affect any first-token evaluation on numeric or unit-prefixed answers.

8 Limitations

The study covers one model and 28 relations. The relationship is correlational: we show that a relation-agnostic readout predicts LRE faithfulness, not that the representation it reads is what the LRE exploits.

The most informative next step is to move the analysis *within* relations. Our unit is currently the relation, so the evidence is 28 aggregate points, and a sceptic may reasonably suspect that both measures track some coarse property of a relation rather than anything about individual entities. The same two quantities are, however, defined per subject: whether the LRE reproduces the model's answer

for that subject, and where the lens ranks it. If the lens predicts *which subjects within a relation* the LRE gets right, that would hold the relation, its prompt, its answer space and its cardinality fixed by construction – controlling by design for everything we currently partial out – and would raise the evidence from 28 points to several hundred. It would also convert the claim from a statement about relations into one about entities, which is what the account actually asserts. Should the lens predict between relations but not within them, the mechanism is coarser than we propose, and that too would be worth knowing.

The account has a scope condition we can state but not yet explain (§6). It is weakest where the answer space is very small, since an LRE can then be faithful by elimination rather than by reading anything from the subject, and where the answer space is very large and each answer near-unique, since faithfulness is then close to its floor regardless. We show that these are the two tails of a single variable, but we do not have a principled account of where the boundaries lie, and with 28 relations we cannot fit one.

We also probe salience through a single neutral prompt, `"{} is known for"`. That phrasing asks what an entity is *noted* for, which need not coincide with everything its representation carries; attributes such as gender may be well encoded without being something a name is "known for". A different neutral prompt, or an average over several, might track salience more broadly.

Seven of the 28 relations have fewer than ten test items after the in-context and reference subjects are removed, so their faithfulness estimates are coarse (for three of them, quantized to thirds); a rotating-split protocol would sharpen these. Finally, the diagnostic is defined only for relations whose subjects are entities, which excludes 19 of the dataset's 47 relations.

9 Conclusion

Whether a relation is linearly decodable from a language model is substantially predictable without the relation. A Jacobian lens estimated from subject differences under a neutral prompt, at a fixed layer, ranks the model's own answer highly for exactly those relations whose LREs turn out to be faithful.

The reading we favour is that salience is the mechanism rather than a confound: a relation admits a faithful linear operator to the extent that its answer is already part of what the subject's representation makes the model disposed to say. On that view the wide spread of LRE faithfulness across relations is less a fact about the relations than about the entities they are asked about – and the exceptions are informative, falling at the two ends of the answer-space range where faithfulness can be achieved, or prevented, without the subject's representation mattering much either way.

References

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Evan Hernandez, Arnab Sen Sharma, Tal Haklay, Kevin Meng, Martin Wattenberg, Jacob Andreas, Yonatan Belinkov, and David Bau. 2024. Linearity of relation decoding in transformer language models. In *Proceedings of the Twelfth International Conference on Learning Representations (ICLR)*.

A All relations

Table 1 gives faithfulness and the lens score for every relation in the study.

relation	k	faith.	A_r	relation	k	faith.	A_r
country_language*	1.7	1.00	-1.04	presidents_election_year†	1.1	0.67	-4.17
country_largest_city*	1.0	1.00	-0.81	product_by_company	17.4	0.66	-3.24
landmark_on_continent	189.4	0.94	-2.81	person_plays_position_in_sport	68.0	0.64	-3.99
person_native_language	30.6	0.94	-2.56	person_band_lead_singer*	1.0	0.60	-1.68
name_birthplace	3.9	0.93	-1.29	person_university	1.3	0.59	-4.52
name_religion	6.2	0.93	-1.37	company_hq	4.1	0.52	-4.09
person_plays_pro_sport	63.6	0.92	-1.45	superhero_person	1.0	0.48	-2.15
person_plays_instrument	85.5	0.88	-1.77	star_constellation	11.7	0.42	-3.67
landmark_in_country	9.2	0.84	-2.25	pokemon_evolutions	1.0	0.39	-3.44
city_in_country	1.3	0.82	-0.48	country_currency	1.3	0.36	-4.48
person_occupation	26.5	0.76	-2.81	superhero_archnemesis	1.1	0.25	-3.57
country_capital_city*	1.0	0.75	-0.74	person_father	1.0	0.09	-3.69
name_gender*	9.5	0.67	-4.67	company_ceo	1.0	0.07	-4.59
presidents_birth_year†	1.3	0.67	-4.24	person_mother	1.0	0.07	-3.99

Table 1: All 28 relations, sorted by LRE faithfulness. k is the mean number of subjects per distinct object; A_r is the relation-agnostic lens score, higher meaning closer agreement with the model. †rescored for the numeric-object artifact (§7); *fewer than ten test items, so faithfulness is coarsely quantized.